Supplementary Materials

Evidence-Retaining Functional Calibration for Few-Trial Motor Imagery EEG Decoding

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S1. Statistical families and data hierarchy

This supplement preserves the complete archived primary BA family and the additional exploratory mechanism family. Participants are the inference units within each cohort. Sessions are averaged within seed and participant, then seeds within participant. The 11/14/10 participant counts are not replaced by the 315/84/30 session–seed counts. Confidence intervals are pointwise percentile participant-bootstrap intervals; their nominal coverage is not multiplicity-adjusted. Primary tests use the archived analysis seed20261026; the new 18-test family uses20261029. Both use10,000 bootstrap draws and exact two-sided sign flips.

Table S1. All12 archived primary BA comparisons. Effects are in percentage points.

Cohort	Comparison	n	ΔBA (pp)	95% CI (pp)	Exact p	Holm p (12)
holdout	ERFC vs ea	11	0.04138	[-0.017342650050870937, 0.10489067071466282]	0.23438	1.00000
holdout	ERFC vs spatial_sgd100	11	0.29125	[-0.2844995699039521, 0.9010806706776812]	0.39160	1.00000
holdout	ERFC vs r9_ea_anchored	11	0.05669	[-0.1134798600472176, 0.23847174351485562]	0.56543	1.00000
holdout	ERFC vs matched_map	11	-0.03830	[-0.16256481464050654, 0.08633799346302315]	0.57617	1.00000
pan2023	ERFC vs ea	14	0.01190	[-0.09523809523809586, 0.15476190476190438]	1.00000	1.00000
pan2023	ERFC vs spatial_sgd100	14	0.23810	[-0.6190476190476191, 0.9880952380952348]	0.60217	1.00000
pan2023	ERFC vs r9_ea_anchored	14	-0.27381	[-0.5595238095238145, 0.011904761904759007]	0.11816	1.00000
pan2023	ERFC vs matched_map	14	-0.27381	[-0.5595238095238121, -0.023809523809522774]	0.09497	1.00000
weibo2014	ERFC vs ea	10	0.02381	[-0.07142857142857117, 0.14285714285714235]	1.00000	1.00000
weibo2014	ERFC vs spatial_sgd100	10	-0.65476	[-2.3095238095238124, 0.5714285714285683]	0.51562	1.00000

Cohort	Comparison	n	ΔBA (pp)	95% CI (pp)	Exact p	Holm p (12)
weibo2014	ERFC vs r9_ea_anchored	10	-0.59921	[-1.3571428571428579, 1.1102230246251565e-15]	0.13281	1.00000
weibo2014	ERFC vs matched_map	10	-0.58333	[-1.0476190476190494, -0.1547619047619131]	0.04688	0.56250

Table S2. All18 newly conducted exploratory matched-MAP tests. Fraction and BA effects use percentage points, whereas NLL uses nats. Holm adjustment covers all18 rows.

Cohort	Outcome	n	ERFC − MAP	95% CI	Exact p	Holm p (18)
holdout	BA (pp)	11	-0.03830	[-0.16197, +0.08572]	0.57617	0.57617
holdout	Balanced NLL (nats)	11	0.00045	[+0.00010, +0.00071]	0.03418	0.17090
holdout	Introduced errors (pp)	11	-0.74658	[-0.86104, -0.63952]	0.00098	0.01367
holdout	Corrected errors (pp)	11	-0.78488	[-0.93173, -0.62772]	0.00098	0.01367
holdout	Probability updates (pp)	11	-83.94967	[-86.79386, -80.76922]	0.00098	0.01367
holdout	Negative-transfer sessions (pp)	11	-27.03660	[-32.86108, -21.33018]	0.00098	0.01367
pan2023	BA (pp)	14	-0.27381	[-0.55952, -0.01190]	0.09497	0.28491
pan2023	Balanced NLL (nats)	14	0.00162	[+0.00087, +0.00243]	0.00049	0.00732
pan2023	Introduced errors (pp)	14	-0.71429	[-0.91667, -0.51190]	0.00024	0.00391
pan2023	Corrected errors (pp)	14	-0.98810	[-1.16667, -0.80952]	0.00012	0.00220
pan2023	Probability updates (pp)	14	-84.79762	[-91.22619, -77.85714]	0.00012	0.00220
pan2023	Negative-transfer sessions (pp)	14	-17.85714	[-26.19048, -9.52381]	0.00391	0.03125
weibo2014	BA (pp)	10	-0.58333	[-1.04762, -0.16667]	0.04688	0.18750
weibo2014	Balanced NLL (nats)	10	0.00347	[+0.00092, +0.00756]	0.00977	0.05859
weibo2014	Introduced errors (pp)	10	-0.69048	[-1.00000, -0.38095]	0.00391	0.03125

Cohort	Outcome	n	ERFC − MAP	95% CI	Exact p	Holm p (18)
weibo2014	Corrected errors (pp)	10	-1.27381	[-1.75000, -0.84524]	0.00195	0.01953
weibo2014	Probability updates (pp)	10	-84.82540	[-90.03968, -79.44444]	0.00195	0.01953
weibo2014	Negative-transfer sessions (pp)	10	-13.33333	[-30.00000, +0.00000]	0.25000	0.50000

S2. Computational scope

Table S3. B20 component timings. Residual fitting starts after feature extraction. Query timing covers additional residual/profile evaluation per1000 already encoded queries. These times exclude support-EA covariance estimation, neural encoding and IO. All18 methods and five budgets are retained at full precision in r10_method_budget_summaries_verified.csv.

Cohort	Method	Residual fit (ms)	Incremental prediction (s/1000)
holdout	profile	1.7458	1.5843
holdout	profile_map	1.8380	0.0536
holdout	r9_ea_anchored	20.3397	0.1178
holdout	support_ea	3.3192	0.0009
pan2023	profile	1.0722	1.5565
pan2023	profile_map	1.7047	0.0479
pan2023	r9_ea_anchored	18.2765	0.1034
pan2023	support_ea	3.2545	0.0009
weibo2014	profile	1.0965	1.6642
weibo2014	profile_map	1.8016	0.0467
weibo2014	r9_ea_anchored	16.1421	0.3347
weibo2014	support_ea	2.6180	0.0005

S3. Implemented configuration and comparators

Table S4. Recorded settings and method-name mapping. “ERFC” is the manuscript name for the unchanged archived profile algorithm, not a new data run. Kernel amplitude and retained-gain allowance have different roles.

Parameter	Value
Source seeds	20261018, 20261019, 20261020
Selected source epochs	39, 30, 45
Input channels × samples	11 × 128
Band-pass / sampling	8–30 Hz / 128 Hz
Task-relative crop / filter context	0.25–1.25 s / 0–1.5 s
Source feature dimension	64
Reference group / sessions	40 participants / 396 sessions
Kernel / amplitude	Linear plus intercept / 0.3
Regularization λ / allowance c	1 / 0.1
UID nugget / probability floor	10^{-10} / 10^{-12}
Newton / bracket / root budgets	60 / 32 / 12

Parameter	Value
Probability-update threshold	Absolute change $\geq 10^{-4}$

Archived implementation	Displayed meaning
frozen	Frozen source decoder
support_ea	Support-only alignment anchor
head_bias	Bias-only head
head_rank4	Rank-four head
head_full	Full head
sgd10	10-step spatial SGD
sgd100	100-step spatial SGD
r3_protected	Protected tangent predecessor
r4_xfit_select_stack	Cross-fitted selection/stacking
r8_fiber_bounded	Bounded geometric predecessor
r9_raw_anchored	Raw-anchor kernel residual
r9_ea_anchored	EA-anchor kernel residual
profile	ERFC
profile_map	Matched MAP residual
profile_no_completion	ERFC without null completion
profile_frozen_anchor	ERFC with frozen-source anchor
profile_quadratic	Quadratic profile approximation
profile_variance_gate	Variance-gated residual

The source checkpoint and input-contract hashes are retained in evidence/source-f0-receipts-proofs.json, evidence/main_method_protocol_verified.json and the source manifests. The source-only selected epochs are39,30 and45 for the three seeds, rather than the maximum candidate length80. The exact channel order is FC3, FC4, C5, C3, C1, Cz, C2, C4, C6, CP3, CP4. At smaller budgets, quality-rejected positions are not replenished. In Stieger, the filter context is anchored to feedback onset2s after the cue. Pan/Weibo retain public-release trial order rather than an asserted verified physical chronology.

S4. Related development evidence

These analyses document the development context and are not independent replications of ERFC. The earlier geometry contrast concerns a different method pair; its four-percentage-point improvement is not substituted for the main ERFC-versus-alignment contrast. Cohorts were reused during development. Interrupted source experiments are not treated as completed confirmatory endpoints.

Table S5. Scope of related development analyses. Full restored source tables are included in tables/ where available.

Analysis	Scope	Result
Protected vs unprotected tangent calibration	Earlier exploratory Stieger comparison; n=11	ΔBA +4.6589 pp; 95% CI [2.3965, 6.9094]; Holm p=0.0439 (10-test family)
Spatiotemporal adaptation evaluation	Repeated-development Pan cohort; n=14	12 primary BA comparisons; none positive after original 24-test multi-metric adjustment

Analysis	Scope	Result
Bounded geometric adaptation	Repeated-development cohorts	40 primary comparisons; none passed Holm adjustment
Anchored residual development	Repeated-development cohorts	24 primary comparisons; none passed Holm adjustment
Controlled mechanism pilot	Eight training-side pilot participants	Six pilot comparisons; none passed Holm adjustment; predeclared stop condition triggered
Spatial-persistence pilot	Same training-side pilot group; n=8	Primary NLL interaction 5.64×10^{-6}; 95% CI $[-2.37\times10^{-5}, 3.33\times10^{-5}]$; exact p=0.703125

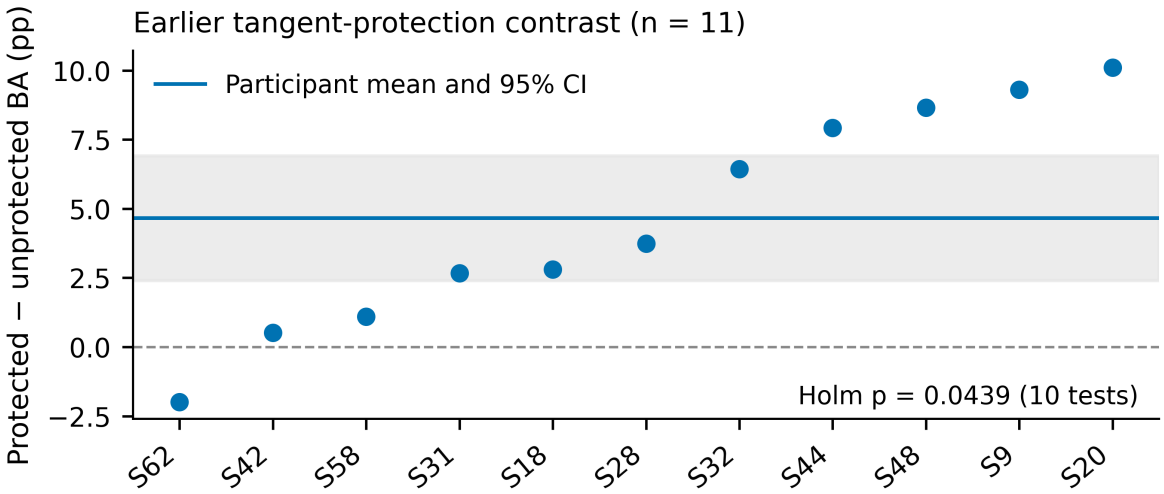


Figure S1. Earlier exploratory protected-tangent comparison. Each point is one participant. The mean is+4.6589 percentage points with the original pointwise95% interval[2.3965,6.9094]; the original20,000-draw participant bootstrap and exact sign-flip analysis yielded Holm-adjusted p=0.0439 in a10-test family. This historical contrast is neither ERFC's main accuracy result nor an independent confirmation.

S5. Reproducibility and provenance

Table S6. Reproduction inputs and operations.

Item	File/operation
Primary archived report and participant summaries	evidence/r10/reports/report.json and participant_metrics.csv
Restored-source hash ledger	evidence/main_download_proofs.json
Original primary comparison family	tables/r10_primary_comparisons_verified.csv
Additional paired effects and independent units	tables/r10_exploratory_matched_map_mechanisms.csv and r10_exploratory_paired_values.csv
Paired mechanism reproduction	python research/analyze_main.py
Original data-derived figures	python figures/plot_figures.py
Main manuscript assembly	python assemble_manuscript.py
Word/PDF rendering	python build_documents.py

The plotting program uses Matplotlib3.10.8 and reads the numerical tables. It produces editable-text SVG, vector PDF,600dpi RGB PNG and LZW-compressed TIFF. The workflow is an original schematic; no generated EEG traces or invented participants are drawn. Source licenses apply to the raw EEG repositories. The reproducibility package supports numerical reanalysis and figure regeneration; rerunning source-model training additionally requires the original EEG data and research code.